



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Understanding Basic Types and What Dynamic Typing and Static Typing are

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Learning Outcome:

Understand key information about the concepts, principles, and methodologies in simple spoken and written situations. Talk about programming techniques, use cases, kinds, and sequences in simple spoken and written situations.

Can Do Statements:

- I can read about basic types.
- I can read and write about dynamic and static typing.
- I can write meaningful and correct sentences to summarize the information related to basic types.

Ask your partner:

- Do you use numbers or letters when programming? Why?
- Why do you think we need to learn about different data types when using a programming language?





Let's Explore!

Marianne, you should consider studying more before the exam.



Instructions: Read the text below.

- Marianne:** Hey Robert! I have an exam tomorrow for the declarative computation model class, and I need to understand basic types and dynamic and static typing. Do you have to go somewhere now or can you help me?
- Robert:** I can help you. You don't have to panic, these topics are easy. First of all, you need to know that a basic data type is a basic building block that we use in programming languages. I suggest reading the programming book to get to know all the basic data types.
- Marianne:** Yes! I'm reading it now, but there's a lot of information and I don't understand everything thoroughly.
- Robert:** Do you need me to help you go over the basic types?
- Marianne:** Let me tell you what I know first. The first type is numbers. They can be integer or floating numbers. Then, we have Booleans that can be either true or false. I know there's also records that are compound data structures. This means that records are a label followed by a group of features like atoms, integers or Booleans. That's all I remember, sorry.
- Robert:** You don't have to worry. I can continue explaining. You also need to learn about strings which are lists of character codes. You can write them in double quotes. We also have procedures that are the values of the procedure type. For this type, we declare a new procedure and then bind it to an identifier. I admit not knowing all of the basic data types, but I think these are the ones you need to keep in mind.
- Marianne:** Thank you! I'm taking notes of everything. Now, I want to avoid having the wrong idea about dynamic and static typing. Do I need to understand these types thoroughly before I start programming? I know this is your specialty.
- Robert:** Not necessarily. Dynamic and static typing only refer to the way a programming language works. Static typed programming languages do not need to define their variables before being used. You only have to declare a variable before you employ it. On the other hand, in dynamic typed programming languages we have to define variables before we use them. That's all.
- Marianne:** Thank you so much for your help!

**Let's Practice!**

Robert is checking information about basic data types.

Instructions: Circle the correct answer in order to complete the sentences. c

1. Don't you (have to / has to) finish that for today, too?
2. What do you (wants to / want to) include specifically?
3. I am looking for information about dynamic and static typing because I (need to / needs to) write the final paper.
4. I don't (having / have to) write the whole report on Basic Data Types.

**Let's Practice!**

Angelina's summary is about dynamic and static typing.

Instructions: Complete the paragraph by filling the blanks. Use: I'm only declaring, I'm reading, are not, have.

_____ about dynamic and static typing. They _____ as difficult as I thought. Static typed programming languages don't need to define their variables before being used. That's why _____ variables when I employ them. On the other hand, in dynamic typed programming languages we _____ to define variables before we use them. That's the main difference between dynamic and static typing.



Extension Activity

Let's write about basic types.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for additional information about basic types.
- Write a paragraph about it using the appropriate grammar.
- Share the paragraph with your classmates.

Self-Assessment

- I look for additional information about basic types.
- I write a paragraph about basic types using the correct grammar.
- I talk about basic types in front of my classmates.

Yes

No

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Target Words

- **building block** (noun): an essential element that serves as the base for more complex elements within the structure
e.g. You need to know that a basic data type is a basic building block that we use in programming languages.
- **thoroughly** (adverb): in a very careful or detailed way
e.g. I'm reading it now, but there's a lot of information and I don't understand everything thoroughly.
- **integer** (noun): a positive or negative number, or zero
e.g. The first type is numbers. They can be integer or floating numbers.
- **keep in mind** (expression): to consider something and take it into account for a specific purpose.
e.g. I admit not knowing all of the basic data types, but I think these are the ones you need to keep in mind.
- **declare a variable** (expression): to indicate that a given variable has a specific name and type.
e.g. Static typed programming languages do not need to define their variables before being used. You only have to declare a variable before you employ it.



Learning Points

Verbs With Infinitives in Negative and Interrogative Forms: Have to and Need to

You can use a **verb** after an **infinitive** to express a purpose.

Infinitive verbs have two characteristics:

- The word **to** goes before the base form of the verb.
- We do not conjugate infinitive verbs. This means that we do not add *-s*, *-es*, *-ed* or *-ing* at the end.

Take a look at the following sentences and pay attention to the infinitive verbs to find, to display and to send.

Examples:

*He doesn't have **to send** the information.*

*I need **to find** this information now.*

*Do you need **to read** this book?*

We can use infinitive verbs after other verbs to indicate a purpose or something that has to be done. There are many verbs that are followed by infinitives and two of the most common ones are the verbs '**have**' and '**need**'. We use these verbs to indicate obligation or strong advice.

This is the structure you use in affirmative sentences:

Subject + **verb (have/need) + infinitive** + complement

Examples:

*We **have to look for** more information about basic types.*

*I **need to read** this article about dynamic and static typing.*

For the **negative** and **interrogative** forms, we have to use the auxiliary verb **to do**.

Let's take a look at the following chart:

Subject Pronouns	Negative	Interrogative	Interrogative with 'wh' word
I / You / We / They	Structure: Subject + do not (don't) + have/need + main verb in infinitive form + complement Example: <i>We don't need to send that email.</i>	Structure: Do + subject + have/need + main verb in infinitive form + complement+? Example: <i>Do they have to sell those products?</i>	Structure: Wh word + do + subject + have/need + main verb in infinitive form + complement + ? Example: <i>When do they have to sell those products?</i>



Learning Points

He / She / It	Structure: Subject + does not (doesn't) + have/need + main verb in infinitive form + complement Example: <i>She doesn't have to buy software.</i>	Structure: Does + subject + have/need + main verb in infinitive form + complement+? Example: <i>Does <i>they</i> have to buy a computer?</i>	Structure: Wh word + does + subject + have/need + main verb in infinitive form + complement + ? Example: <i>Why <i>does</i> <i>she</i> have to buy a computer?</i>
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Examples

- *Do you have to go somewhere now or can you help me?*
- *You don't have to worry. I can continue explaining.*
- *Do you need me to help you go over the basic types?*
- *Static typed programming languages do not need to define their variables before being used.*

Gerunds After Certain Verbs: Admit, Avoid, Continue

Some verbs are followed by a gerund which is the **ing** form of a verb.

When we use two verbs in the same sentence, sometimes, the second verb takes its **ing form**. There are some specific verbs that are followed by gerunds. The most common verbs followed by a gerund are these: *avoid, continue, admit, enjoy, consider, miss, imagine, suggest* and *like*.

Examples:

*I continue practicing **ing** English.*

*She enjoys working **ing** here.*

*Jack avoids sending **ing** more information.*

Examples

- *I suggest reading the programming book to get to know all the basic data types.*
- *You don't have to worry. I can continue explaining.*
- *I admit not knowing all of the basic data types, but I think these are the ones you need to keep in mind.*
- *I want to avoid having the wrong idea about dynamic and static typing.*





Learning Points

Word Formation of Nouns: Suffixes -tion and -ity

A **suffix** is a group of letters located at the end of a word. A suffix can make a new word in one of two ways:

- Inflectional (grammatical): for example, changing singular to plural (dog → dogs), or changing present tense to past tense (walk → walked).
- Derivational (the new word has a new meaning, "derived" from the original word): for example, teach → teacher

Noun suffixes: tion

When you add a suffix to a **verb form** you can create a **noun**.

Examples:

Transport- transportation

Create- creation

Invent- invention

Inform- information

Noun suffixes: ity

When you add a suffix to an **adjective** you can create a **noun**.

Examples:

Absurd -absurdity

Complex- complexity

Examples

- *I'm reading it now, but there's a lot of information and I don't understand everything thoroughly.*
- *I want to avoid having the wrong idea about dynamic and static typing. Do I need to understand these types thoroughly before I start programming? I know this is your speciality.*